Ravi Pandey

Software Engineer

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Summary

Software engineer building backend services, web applications, and data pipelines with Python, TypeScript, and SQL. Experience in REST API integrations, automated testing, and CI/CD, with graduate research in compiler development.

Experience

CIST Graduate Research Assistant **Jan 2025–Present**
University of Nebraska Omaha, Omaha, NE *Part-time, 20 hours/week*

- Built **FunctionalASP**, an Answer Set Programming extension with user-defined functions and aggregates; implemented its Python compiler, parser and AST transformations, and Clingo backend integration.
- Configured **GitHub Actions CI** with nox for strict mypy checks, unit and integration tests, coverage enforcement, and formatting; benchmarked compiler transformations on **180 ASP Competition instances**.
- Developed a **NeurASP inference system** using Clingo and reusable **PyTorch modules** for differentiable reasoning; implemented a separate splitting prototype with dependency analysis and cached state transitions.

Associate Software Engineer **Jun 2023–Aug 2024**
GrowByData Services, Lalitpur, Nepal

- Developed **Python/Pandas ETL pipelines and API integrations** for advertising and product data, handling request failures, schema changes, and malformed responses to produce reusable client reports.
- Worked with stakeholders on reporting requirements and delivered **Angular/TypeScript analytics features** backed by REST APIs; resolved production tracking and reporting discrepancies.

Selected Projects

FarmAI GitHub | Web Interface **2026**

- Built **FastAPI REST endpoints** and a **React/TypeScript** review interface for scanned records, with a separate background worker, SQLite job persistence, progress tracking, corrections, and CSV/JSON export.
- Implemented **JSON parsing, validation, retries, and resumable checkpoints** for vision-LLM OCR with template-aware prompts; supported **9 form layouts** and evaluated outputs against **712 ground-truth cells**.
- Deployed on **Ubuntu** with systemd and Apache; built **GitHub Actions CI/CD** that checks Python and frontend changes before deployment, with health checks, SQLite backups, and code rollback in the release script.

Agentic Harness GitHub **2026**

- Built a **Python/LangGraph** workflow coordinating Research, Developer, QA, and Release agents, with **JSON Schema** output contracts, Pydantic validation, bounded repair loops, and saved run artifacts.
- Implemented automated QA and **pytest** checks for generated Python code in restricted **Docker** containers; required human approval before staging output or creating a draft GitHub pull request.
- In a recorded GitHub connector demo, one revision resolved two failing tests, resulting in **14 passing tests** and a draft pull request after approval.

XAI Medical Recommender GitHub **2025**

- Built a **Python/Streamlit** recommendation interface with streaming LLM explanations, constructing prompts from symptoms, model predictions, and session context.
- Designed **SQL tables and Supabase logging** for conversations, surveys, and quizzes; built a companion quiz workflow and evaluated user understanding in a **20-participant study**.

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, Bash
Backend & Data: FastAPI, REST APIs, SQLite, MySQL, Supabase, Pandas, JSON Schema, Pydantic
Frameworks & Libraries: React, Angular, Streamlit, OpenCV, LangGraph
Testing & Delivery: Git, GitHub Actions CI/CD, Docker, Linux, pytest, unittest, nox, mypy
Cloud Familiarity: AWS (S3, Lambda, SQS), Azure

Education

University of Nebraska Omaha **Dec 2026 (Expected)**
M.S. Computer Science GPA: **4.0**

Pulchowk Campus, Tribhuvan University **2018–2023**
B.E. Computer Engineering GPA: **3.73**

Selected Recognition

$5,000 GRACA research grant, University of Nebraska Omaha | **First Prize**, openIMIS-DRG Datathon 2022